

Software Requirements Specification

for

Online Internship & Placement Preparation Portal

Version 1.0

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Revision History

Name	Date	Reason for changes	Version
v1.0	Sep 2, 2026	Initial SRS draft	1.0

1. Introduction

1.1 Purpose This document specifies the complete software requirements for Version 1.0 of the Online Internship & Placement Preparation Portal. It defines all functional and non-functional requirements for the entire system, covering student/admin account management, internship listing and application handling, placement test practice, and performance-based material recommendation. This SRS covers the full system; no subsystem is excluded.

1.2 Intended Audience This document is intended for the development team (Samyuktha, Shreya, Sharvane, Sanand) and the course faculty/evaluators reviewing the project. Readers new to the project should start with Section 2 for a high-level overview, then read Section 5 for detailed feature requirements relevant to their assigned module, and Section 6 for quality expectations.

1.3 Product Scope The Online Internship & Placement Preparation Portal is a web-based platform built for university engineering placement cells. It replaces manual, spreadsheet- and email-based coordination between placement officers and students with a centralized system for posting internships, managing applications, and preparing students through domain-based testing and performance analysis. The product's goal is to reduce administrative overhead for placement officers and improve student readiness by tailoring practice material to individual weak areas identified through test performance.

1.4 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications (template basis for this document)
- Project proposal document, Group 15, "Online Internship & Placement Preparation Portal"

2. Overall Description

2.1 Product Perspective This is a new, self-contained web application. It is not a follow-on to an existing system. It consists of a single web application with two role-based views (Student and Admin) sharing one backend and database, so there are no external subsystem interfaces to define beyond standard client-server communication.

2.2 Product Functions

- Account registration and profile management for students and admins
- Internship listing creation, editing, and deletion (admin) and browsing/filtering (student and admin)
- Internship application submission with eligibility validation
- Application status tracking (student) and application review/filtering (admin)
- Question and test bank management (admin)
- Test attempt, submission, and automatic scoring (student)
- Performance analysis (strengths/weaknesses) and progress tracking
- Personalized preparation material recommendation based on weak areas

2.3 User Classes and Characteristics

- **Student:** Engineering undergraduate seeking internships. Uses the system frequently to browse internships, apply, take tests, and check recommendations. Basic-to-moderate technical proficiency assumed; no admin privileges.
- **Admin (Placement Officer):** Manages internship postings, reviews applications, and curates the question bank. Uses the system regularly but performs fewer, higher-privilege actions. Requires authenticated elevated access.
- **Head Admin:** The same as a regular admin but also has the privilege of approval of admin account creation requests.

2.4 Operating Environment

- Client: Modern web browsers (Chrome, Firefox, Edge, Safari — latest two versions), responsive design for desktop.
- Server: Cloud computing provider, ex. AWS or Vercel.
- Database: MySQL (relational, given structured entities like students, internships, applications, questions, results).
- Frontend & backend: Next.js single-page application.

2.5 Design and Implementation Constraints

- Resume uploads restricted to PDF format, max 5 MB.
- Role-based access control (RBAC) enforced at the API layer by making students and admins hit different endpoints.
- All passwords stored using hashed encryption (e.g., bcrypt); no plaintext storage.

2.6 Assumptions and Dependencies

- Assumes users have consistent internet access while using the portal.
- Assumes the university provides accurate academic records for eligibility validation (self-reported by students at registration).
- Depends on availability of chosen hosting provider and MySQL database service.
- Assumes no integration with external Applicant Tracking Systems (ATS) or company portals in v1.0.

3. External Interface Requirements

3.1 User Interfaces

- **Login/Registration screen:** registration and login, form fields, and validation messages. Administrator accounts shall be created or managed only through an authorized administrative process via a head admin.
- **Student Dashboard:** profile summary, internship listings (with filters), application status tracker, test/practice section, recommendations panel.
- **Admin Dashboard:** internship management panel (add/edit/delete), applicant review panel, question bank management panel, student profile search.
- **Admin Account Approval Dashboard:** Visible only to head admins, this page will show all requests for admin account creation as well as the option to accept or reject them.

- Common UI elements across all screens: top navigation bar, search/filter bar wherever listings appear, consistent color-coded status tags (e.g., green = Selected, yellow = Pending, red = Rejected), standardized error/success toast messages.

3.2 Software Interfaces

- **Database:** MySQL 8.x — stores Users, Students, Admins, Internships, Applications, Questions, Tests, Results, and Recommendations tables.
- **Full-stack Framework:** Next.js, having both the frontend code as well as the backend API configuration.
- No third-party software integrations (e.g., resume parsers, external APIs) are included in v1.0.

3.3 Communications Interfaces

- All client-server communication occurs over HTTPS.
- REST API using JSON as the data interchange format.
- Optional email notifications (via SMTP, e.g., using Nodemailer) for application status changes. To be implemented if time permits; not mandatory for v1.0.

4. Analysis Models

Use Case Diagram (described): Actors: Student, Admin, Head Admin.

- Student use cases: Register/Login, Manage Profile, Browse Internships, Apply to Internship, Track Application Status, Attempt Test, View Results, View Recommendations.
- Admin use cases: Login, Manage Internship Listings, Review Applications, Manage Question Bank, Search/Filter Student Profiles.
- Head Admin use cases: Everything a regular admin does but includes approval or rejection of admin account creation requests.

5. System Features

5.1 Account & Internship Listing Management (Owner: Samyuktha Ramesh Babu)

5.1.1 Description and Priority: This feature covers account creation and profile maintenance for both students and admins, as well as posting and browsing internship listings. Priority: High.

5.1.2 Stimulus/Response Sequences:

When a student registers, they provide their personal and academic details and can optionally upload a resume before the system creates their student account. Once logged in, a student can access their profile, update their skills or academic information, and save the changes. Administrators can log in through an authorized administrative account and search or filter student profiles to view relevant details. Separately, an administrator can add a new internship by entering the company, role, domain, location, required skills, eligibility criteria, and deadline, after which the internship becomes visible to students. Students and administrators can both

apply filters on the internship listings page to narrow down results by domain, location, skills, or deadline.

5.1.3 Functional Requirements:

- REQ-1: The system shall allow students to create an account by providing their name, email, password, and academic details (branch, CGPA, and year), along with an optional resume upload in PDF format (maximum 5 MB). The system shall reject registration with a clear error message if mandatory fields are missing or the email is already registered.
- REQ-2: On registration, the system shall allow admin or head admin account creation by providing name, email, and password. Once submitted, the person may log in and view the status of their request (pending, accepted, rejected). Once the result of the request has come in (accepted, rejected), on acknowledgement (ex. pressing an "OK" button), the account may be (1) deleted on rejection (credentials will become invalid), or (2) turned into a regular admin account. The system shall reject registration with a clear error message if mandatory fields are missing or the email is already registered.
- REQ-3: Students and admins can delete their own accounts, but at any given point, there must be at least one admin account designated as a head admin.
- REQ-4: The system shall allow head admins to view all admin account creation requests on a single page, and accept or reject them.
- REQ-5: The system shall allow students to view and update their profile, skills list, and academic information, and shall allow admins to search and filter student profiles by name, branch, CGPA range, and skills.
- REQ-6: The system shall allow admins to create, update, and delete internship listings with company, role, domain, location, required skills, eligibility criteria, and application deadline, while students shall only be able to view listing details.
- REQ-7: The system shall allow both students and admins to filter internship listings by company, role, domain, location, skills, and deadline.
- REQ-8: The system shall automatically mark internship listings as "closed" when the application deadline passes.

5.2 Internship Application Management (Owner: Sanand Kumar Shetty)

5.2.1 Description and Priority: This feature lets students apply to internships and track their application status, while giving admins the ability to view applications. Priority: High.

5.2.2 Stimulus/Response Sequences:

When a student clicks "Apply" on a listing, the system checks whether they meet the eligibility criteria. If they're eligible, the application gets submitted with status "Pending"; if not, the system shows a message explaining why and blocks the submission. On the admin side, an admin can open the Applications panel, filter by internship, student, or status, and update the status of any application after reviewing it.

5.2.3 Functional Requirements:

- REQ-9: The system shall allow students to submit an application for any internship listing they are eligible for.
- REQ-10: The system shall validate a student's eligibility (CGPA, skills, branch) against the internship's criteria before allowing submission, and shall block ineligible applications with an explanatory message.

- REQ-11: The system shall allow students to view the real-time status (Pending/Shortlisted/Selected/Rejected) and deadline for each of their submitted applications.
- REQ-12: The system shall allow admins to view and filter all student applications by internship, student name, or status.
- REQ-13: The system shall allow admins to update the status of any submitted application.

5.3 Placement Test & Practice (Owner: Shreya Narayana Bhat)

5.3.1 Description and Priority: This feature provides a domain-tagged question bank along with a test-taking interface for students. Priority: High.

5.3.2 Stimulus/Response Sequences:

When an admin adds a new question or test, they tag it with a domain and difficulty level, after which it becomes available in the bank. A student can select a domain, usually linked to an internship they're interested in, attempt the corresponding test, and submit it, at which point the system scores it immediately.

5.3.3 Functional Requirements:

- REQ-14: The system shall allow admins to add, edit, and remove individual questions and grouped tests, each tagged with a domain and difficulty level.
- REQ-15: The system shall allow admins to view, add, edit, delete, and filter questions and tests by domain and difficulty. Students shall only be able to view and answer questions presented to them as part of an assigned or selected test
- REQ-16: The system shall allow students to select a domain for their selected or interested internship and attempt tests relevant to the selected domain.
- REQ-17: The system shall automatically calculate and display the student's score and percentage immediately upon test submission.
- REQ-18: The system shall store each test result, including score, percentage, timestamp, and question-wise correctness, for later analysis, and shall allow students to retake practice tests while maintaining a record of previous attempts.

5.4 Result Analysis & Material Recommendation (Owner: Sharvaneer Naru)

5.4.1 Description and Priority: This feature analyzes a student's test performance to identify their strengths and weaknesses and recommends suitable preparation material. Recommendations are generated using predefined rules based on the student's test performance. Priority: Medium-High.

5.4.2 Stimulus/Response Sequences:

Once a student completes a test, the system updates their domain-wise performance statistics in the background. When the student opens their "My Progress" page, they can see a breakdown of their strong and weak domains along with recommended resources for improvement.

5.4.3 Functional Requirements:

- REQ-19: The system shall analyze a student's test results across attempts to identify their domains of strength and weakness, based on average performance per domain.

- REQ-20: The system shall maintain and display a student's progress trend over time, such as their score history per domain.
- REQ-21: The system shall recommend specific preparation materials targeted at each identified weak domain.
- REQ-22: The system shall automatically update a student's recommendations whenever a new test result is recorded.
- REQ-23: The system shall allow students to view their complete analysis, including strengths, weaknesses, progress, and recommendations, on a dedicated dashboard page.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

- Page load time shall not exceed 4 seconds under normal network conditions.
- Test score calculation and display shall occur within 3 seconds of submission.
- Filter/search operations on listings and question banks shall return results within 3 seconds.

6.2 Safety Requirements Not physically safety-critical.

6.3 Security Requirements

- All passwords shall be stored using a one-way hash (bcrypt).
- Role-based access control shall restrict admin-only actions (listing/question management, application status updates) from student accounts.
- Students shall only be able to access and modify their own profile, applications, and results, not those of other students.
- All communication shall use HTTPS/TLS encryption.

6.4 Software Quality Attributes

- **Usability:** Dashboards shall be intuitive enough for a first-time user to apply to an internship or attempt a test without external guidance.
- **Reliability:** Test scoring and eligibility checks shall be accurate 100% of the time given valid input data.
- **Maintainability:** Codebase shall be modular, with each of the four feature areas (profile & listings, applications, tests, analysis & recommendation) in separate backend modules matching feature ownership.

6.5 Business Rules

- Only admins may create, edit, or delete internship listings and question bank entries.
- Students may apply to an internship only if they meet its stated eligibility criteria.
- Only the student who owns a result/recommendation record may view it; admins may view aggregate/individual student profiles but not raw test-attempt answers unless explicitly permitted.
- Domain requirement: internship domains and question/test domains shall use a shared, predefined domain list (e.g., Web Development, AI/ML, Core CS, Networks, Databases) to ensure consistent filtering and recommendation matching.

7. Other Requirements No additional legal, internationalization, or reuse requirements apply to this project at this stage.

Appendix A: Glossary

- **Admin:** A placement officer with elevated privileges to manage internships, applications, and the question bank.
- **Head admin:** An admin with the privilege to approve admin and head admin account creation requests.
- **Domain:** A subject/skill area (e.g., Web Development, AI/ML) used to categorize internships, questions, and tests.
- **Eligibility criteria:** Minimum requirements (CGPA, branch, skills) a student must meet to apply for a specific internship.
- **RBAC:** Role-Based Access Control — restricting system functions based on user role (Student/Admin).

Appendix B: Field Layouts

All text fields only allow ASCII characters.

Student Registration Fields: Name (text), Email (text, unique), Password (text, hashed), Branch (dropdown), CGPA (decimal), Year (dropdown), Skills (multi-select tags), Resume (PDF upload, optional, must be no more than 5MB in size).

Admin Registration Fields: Name (text), Email (text, unique), Password (text, hashed).

Internship Listing Fields: Company (text), Role (text), Domain (dropdown), Location (text), Skills Required (multi-select tags), Eligibility Criteria (min CGPA, branch), Deadline (date).

Question Fields: Question Text (text), Domain (dropdown), Difficulty (Easy/Medium/Hard), Options (4 text fields), Correct Answer (dropdown).

Report Requirements (Admin export): Student Name, Application Count, Selected Internships, Average Test Score, Weak Domains.

Appendix C: Requirement Traceability Matrix

ID	Name	Design reference	Test cases
REQ-1	Student account creation	1. Go to URL of site 2. Click create student account 3. Enter name, email, password, academic details, and optionally resume. 4. Click create.	TC-01

REQ-2	Admin account creation	1. Go to URL of site 2. Click create admin account 3. Enter name, email, and password. 4. Click create.	TC-02
REQ-3	Account deletion and head admin existence	1. Go to URL of site 2. Log in as student/admin 3. Navigate to account settings 4. Attempt to delete the account.	TC-03
REQ-4	Admin account creation request management	1. Go to URL of site 2. Log in as head admin 3. Navigate to requests page 4. View, accept, or reject an admin account creation request.	TC-04
REQ-5	Student profile updation and student search	1. Go to URL of site 2. Log in as student/admin 3. Navigate to profile or student search page 4. Update profile or search/filter student profiles.	TC-05
REQ-6	Internship listing management	1. Go to URL of site 2. Log in as admin or head admin 3. Navigate to internship listing page 4. Create, update, or delete an internship listing.	TC-06
REQ-7	Internship listing filtering	1. Go to URL of site 2. Log in as any role 3. Navigate to internship listing page 4. Select one or more filtering criteria 5. View filtered results.	TC-07
REQ-8	Internship list item closing	1. Go to URL of site 2. Navigate to internship listing page 3. Open an internship whose deadline has passed 4. Check whether applications are closed.	TC-08
REQ-9	Internship application submission	1. Go to URL of site 2. Log in as an eligible student 3. Select an open internship 4. Click "Apply".	TC-09
REQ-10	Internship eligibility validation	1. Go to URL of site 2. Log in as student 3. Select an internship	TC-10

		4. Attempt to apply with different CGPA, branch, and skills combinations.	
REQ-11	Application status tracking	1. Go to URL of site 2. Log in as student 3. Navigate to application status page 4. View submitted applications, statuses, and deadlines.	TC-11
REQ-12	Admin application viewing and filtering	1. Go to URL of site 2. Log in as admin 3. Navigate to applications page 4. View and filter applications by internship, student, or status.	TC-12
REQ-13	Application status update	1. Go to URL of site 2. Log in as admin 3. Navigate to applications page 4. Select an application 5. Change its status.	TC-13
REQ-14	Question and test management	1. Go to URL of site 2. Log in as admin 3. Navigate to question/test bank 4. Add, edit, or delete questions/tests 5. Specify domain and difficulty.	TC-14
REQ-15	Question/test viewing and filtering	1. Go to URL of site 2. Log in as admin/student 3. Navigate to question/test section 4. Admin views/filters questions/tests; student accesses questions through a selected/assigned test.	TC-15
REQ-16	Domain-specific test attempt	1. Go to URL of site 2. Log in as student 3. Select a domain 4. Select and attempt a relevant test 5. Submit the test.	TC-16
REQ-17	Automatic test score calculation	1. Go to URL of site 2. Log in as student 3. Attempt a test with known correct and incorrect answers 4. Submit the test	TC-17

		5. View score and percentage.	
REQ-18	Test result storage and retakes	1. Go to URL of site 2. Complete a test 3. View the stored result 4. Retake the same test 5. Verify that previous attempts remain recorded.	TC-18
REQ-19	Strength and weakness analysis	1. Go to URL of site 2. Complete tests across multiple domains 3. Navigate to analysis page 4. View identified strengths and weaknesses.	TC-19
REQ-20	Progress trend tracking	1. Go to URL of site 2. Complete multiple tests over time 3. Navigate to progress page 4. View score history/trends per domain.	TC-20
REQ-21	Weak-domain material recommendation	1. Go to URL of site 2. Complete tests resulting in weak domains 3. Navigate to recommendations page 4. View recommended preparation materials.	TC-21
REQ-22	Automatic recommendation update	1. Go to URL of site 2. Record an existing test result 3. Complete another test 4. View recommendations again.	TC-22
REQ-23	Complete analysis and recommendation view	1. Go to URL of site 2. Log in as student 3. Navigate to analysis/recommendation dashboard 4. View strengths, weaknesses, progress, and recommendations.	TC-23

Appendix D: Test case enumeration

ID	Purpose	Input	Pass output	Fail output
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TC-01	Checking student account creation with valid and invalid information.	Valid student registration details, followed by missing/invalid mandatory fields and duplicate email. Optional resume may be provided.	Valid registration creates the account; invalid or duplicate registration is rejected with a clear error message.	Invalid registration is accepted, or valid registration is rejected.
TC-02	Checking admin account creation and request submission.	Valid admin name, email, and password, followed by invalid/missing information or duplicate email.	Valid request is submitted with pending status; invalid/duplicate registration is rejected with a clear error message.	Invalid request is accepted or valid request is rejected.
TC-03	Checking account deletion while ensuring that at least one head admin remains.	Student/admin attempts to delete their own account; only remaining head admin attempts deletion.	Student/admin account can be deleted; deletion of the only head admin is blocked.	Account deletion violates the minimum head admin requirement or valid deletion is blocked.
TC-04	Checking head admin management of admin account creation requests.	Head admin views pending requests, accepts one request, and rejects another.	Requests are displayed correctly; accepted account becomes a regular admin after acknowledgement and rejected account becomes invalid/removed.	Requests cannot be managed correctly, or accepted/rejected accounts have the wrong final state.

TC-05	Checking student profile updates and admin student profile search/filtering.	Student changes valid and invalid profile/academic/skills information; admin searches students using name, branch, CGPA range, and skills.	Valid profile changes are saved and invalid changes are rejected; admin receives correct matching student profiles.	Invalid profile information is saved or search/filter results are incorrect.
TC-06	Checking internship listing create, update, and delete operations.	Admin creates an internship with valid details, modifies its information, and deletes it.	Internship is created, updated, and deleted successfully; resulting listing information is displayed correctly.	One or more of the create, update, or delete operations fails or produces incorrect listing information.
TC-07	Checking internship listing filtering.	Company, role, domain, location, skills, and/or deadline filtering criteria.	Only internships matching the selected criteria are displayed.	Non-matching internships are displayed or matching internships are omitted incorrectly.
TC-08	Checking automatic closure of an internship after its application deadline.	Internship whose application deadline has passed.	Internship is automatically marked as closed and students cannot submit applications.	Internship remains open or accepts applications after the deadline.
TC-09	Checking application submission for an eligible student.	Logged-in student satisfies all eligibility criteria and selects an open internship.	Application is submitted successfully with status "Pending".	Eligible student cannot submit the application or application receives an incorrect initial status.

TC-10	Checking internship eligibility validation.	Students with different CGPA, branch, and required-skill combinations attempt to apply for an internship.	Application is allowed only when all stated eligibility criteria are satisfied; otherwise submission is blocked with an explanatory message.	An ineligible student can submit an application or an eligible student is incorrectly blocked.
TC-11	Checking student application status and deadline tracking.	Student has one or more submitted applications with different statuses and deadlines.	Student can view the current status and deadline for every submitted application.	Status/deadline information is missing, outdated, or incorrect.
TC-12	Checking admin application viewing and filtering.	Admin opens the application panel and filters by internship, student name, and/or status.	All relevant applications are displayed and filters return the correct applications.	Applications are missing or filtering returns incorrect results.
TC-13	Checking application status updates by admins.	Admin changes application statuses between Pending, Shortlisted, Selected, and Rejected.	New status is saved and correctly reflected in the student's application status tracker.	Status is not saved or an incorrect status is displayed.
TC-14	Checking question and test create, update, and delete operations.	Admin adds, edits, and deletes questions/tests with domain and difficulty information.	Questions/tests are created, updated, and deleted successfully with the correct domain and difficulty information.	One or more operations fails or incorrect information is stored.

TC-15	Checking question/test viewing, filtering, and role-based access.	Admin filters questions/tests by domain and difficulty; student attempts to access question/test management functionality.	Admin can view and filter questions/tests; student can only access questions through an assigned/selected test and cannot manage the question bank.	Student can modify/manage questions or admin filtering/viewing does not work correctly.
TC-16	Checking student selection and attempt of a domain-specific test.	Student selects a domain and a test associated with that domain, answers questions, and submits.	Relevant test is presented and can be completed and submitted successfully.	Student receives an unrelated test or cannot complete/submit the selected test.
TC-17	Checking automatic calculation and display of test score and percentage.	Completed test containing a known number of correct and incorrect answers.	System immediately calculates and displays the correct score and percentage.	Score or percentage is incorrect, missing, or not displayed after submission.
TC-18	Checking test result storage and retake functionality.	Student completes and submits a test, views the result, then retakes the same test.	Score, percentage, timestamp, and question-wise correctness are stored for each attempt, and previous attempts remain available.	Result information is not stored correctly or previous attempts are overwritten/lost.
TC-19	Checking identification of strengths and weaknesses.	Student has test results with different average performance across multiple domains.	System correctly identifies strong and weak domains based on average performance.	Strength/weakness analysis does not correspond to the student's stored performance.

TC-20	Checking display of student progress trends.	Student has multiple test attempts across different dates and domains.	Score history/progress trend is displayed correctly for each relevant domain.	Progress history is missing or does not reflect the student's test results.
TC-21	Checking recommendation of preparation material for weak domains.	Student has one or more identified weak domains.	Preparation materials relevant to each identified weak domain are recommended.	Recommendations are missing or unrelated to the identified weak domains.
TC-22	Checking automatic update of recommendations after a new test result.	Student completes a new test that changes their domain-wise performance.	Recommendations are automatically updated based on the new test result.	Recommendations remain unchanged or are based on outdated results.
TC-23	Checking complete student analysis and recommendation dashboard.	Student with recorded test results opens the analysis/recommendation dashboard.	Dashboard displays strengths, weaknesses, progress, and recommendations correctly.	One or more required analysis/recommendation components are missing or incorrect.